

Perceived Transparency in Optical See-Through Augmented Reality

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ABSTRACT

Optical see-through (OST) displays overlay rendered images to the real-world background creating augmented reality (AR). But the blending between the rendering and the real-world background introduces perceived transparency. The increased luminance on the rendering decreases the background contrast, and reduces the perceived transparency, which is not incorporated in existing color appearance models or display color management pipelines. We studied the perceived transparency in AR focusing on the interaction between the rendering and the patterned background in various luminance, contrast, and wave forms. In addition to AR contrast, we also examined simulated contrast modulation by changing the luminance amplitude. Two psychophysical experiments were conducted to quantify the perceived transparency. The first experiment measured a perceived transparency scale using direct scaling, and the second experiment evaluated the transparency equivalency between two methods of contrast modulation. The result showed that the two methods evoke different transparency perceptions. The background contrast affects the perceived AR transparency significantly, while the background luminance and wave form do not. We proposed a model predicting the perceived transparency based on the first experiment result from AR luminance and background contrast. The model was verified with the second experiment and showed good prediction. Our model presented a new perceptual dimension in OST AR and can possibly be incorporated into color management pipeline to improve AR image quality.

Keywords: Augmented Reality, See-through Displays, Contrast, Transparency, Color Appearance.

Index Terms: I.3.7.b [Computing Methodologies]: Computer Graphics -Three-Dimensional Graphics and Realism-Color, shading, shadowing, and texture.

1 INTRODUCTION

Optical see-through (OST) displays in Augmented reality (AR) or mixed reality (MR), offer direct interaction between the reality and the computer-generated virtuality. OST displays overlay rendered images upon the real-world background by projecting light onto a transparent optical combiner. The rendered AR images will have the background blended in, which not only distorts the rendered colors, but also induces transparency sensation (Figure 1). Although transparency in general describes the quality allowing one to see through, its scientific definition has not been universal. In International Lighting Vocabulary by International Commission on Illumination (CIE), perceived transparency is defined as “the degree of visibility of an object through a medium” [1]. We adopted this definition focusing on the transparent media instead of the background. In this study we

define the perceived transparency as the degree of “see-through”-ness of a layer. Translucency and transparency are interchangeable in this context since minimal scattering presents in OST AR.



Figure 1. Transparency in AR: a photograph taken through an AR HMD, in which the AR chair and colorful dice appear transparent.

Perceived transparency in AR is not incorporated in existing color appearance models such as CIECAM02 [2] and CAM16 [3], which predict and manage displayed colors to improve image quality. Effort has been made to improve AR image visibility problem due to the high transparency [4], [5]. But there has been little research on quantifying perceived transparency directly in OST AR, though it is the nature of OST displays and requires thorough understanding to improve AR image quality.

Perceived transparency has been studied on various other media. Early research addressed the simulated transparency illusion where no light was physically transmitted but luminance difference between adjacent areas [6], [7]. The transparency perception was evaluated in transmittance and lightness, and demonstrated to be dependent on the perceived contrast [8], [9]. In subtractive filters the material appearance depends on the physical properties including reflectance, transmittance, absorptance, and scattering [10]. The perceived transparency of subtractive filters was assessed in two dimensions: perceptual transmittance and clarity [11], [12], which correspond to how much light is transmitted, and how cloudy the filter appears. In addition to flat transparent layers, rendered and real voluminal objects were also explored [13], [14].

Perceived transparency in AR is different from simulated or subtractive filter transparency due to the additional light overlaid. Transmittance and clarity in subtractive filter transparency cannot be applied to the AR directly because the latter is not induced by the physical properties. The perceived transparency of AR renderings relates to the luminance of the AR overlay and the background properties. We consider the luminance contrast with neutral monochromatic colors. Michelson contrast C [15] is used to for repetitive patterns like sinusoidal gratings:

$$C = \Delta L / (2\bar{L}) \quad (1)$$

There are two methods to modulate contrast: by varying the pattern average luminance $\bar{L}$, or by varying peak-to-peak luminance difference ΔL . This contrast modulation can be described in a Cartesian space (Figure 2) with the base pattern as the surround. At the origin, the pattern is the same as the base

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pattern. Along the x axis is the change in peak-to-peak luminance difference ΔL , and along the y axis is the change in average luminance $\bar{L}$. In this contrast-modulated-transparency space, we could use the base pattern as the background. Simple additive AR rendering lies along the positive y axis, reducing background contrast and AR rendering transparency by increasing average luminance $\bar{L}$. But with careful registration to the background, it can be extended to quadrant I and II. Physical subtractive filters are in the quadrant II and III depending on the filter reflectance, transmittance, absorbance, and scattering.

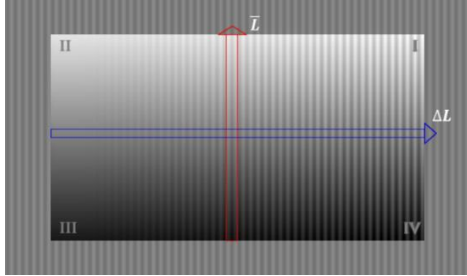


Figure 2. Contrast modulated transparency space: the pattern at the origin is the same as the base pattern in the surround. Pattern luminance amplitude change is along the x axis (blue), and average luminance along the y axis (red). Simple AR overlays are along the positive y axis.

In this study, we examined the contrast modulated AR transparency in relation to different backgrounds with two experiments. First a direct anchored scaling experiment was used to measure the perceived transparency scale at various background luminance and contrast levels. In addition to the AR overlay patches, we also included simulated transparent patches reducing background contrast by reducing luminance amplitude ΔL . This type of patch was tested to verify if contrast is the fundamental mechanism of the transparency modulation. The second experiment was a paired comparison between the two types of patches with simple staircases to identify the point of equivalent transparency. We modeled the perceived AR transparency from the scaling experiment result and verified with the equivalency experiment result.

2 METHOD

2.1 Experiment environment

The OST AR system consisted of an upward facing 27-inch display monitor with resolution of 1920x1080 at the bottom of the box, and a slanted beam splitter as in Figure 3. The AR rendering was generated from the bottom display monitor and reflected by the beam splitter (blue arrow). The 27-inch background display with the same resolution, namely the background LCD, was also visible through the beam splitter (orange arrow) acting as the real-world background. Both the AR rendering and the background are combined and visible through viewing port occupying roughly 25°x13° field of view (FoV). The two displays were set to be at the same optical distance at 110 cm from the observer, so they appeared approximately co-planar at the same depth. This was to minimize the vergence-accommodation conflict and create conformable viewing experiment when switching between the AR and the background. The experiments were conducted in a dark room. Behind the background LCD was the black drop cloth illuminated with D65 with brightest area on the drop cloth visible to the observers were at 5.3 cd/m² to create a more natural viewing condition than completely dark [16].

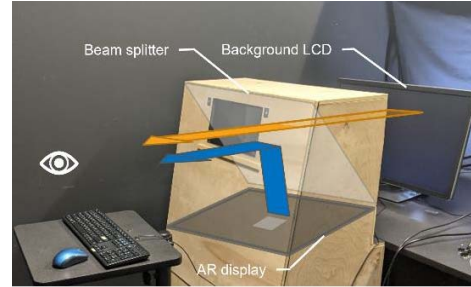


Figure 3. Experiment setup showing a cutaway of the optical blending of the image from the background LCD through the beam splitter (orange arrow) and from the AR display reflected by the beam splitter (blue arrow).

2.2 Stimulus

We selected 6 sinusoidal and 1 square wave gratings as the backgrounds specified in Figure 4 covering as much of the luminance and contrast range of our apparatus. A square wave was included to examine if the high frequency edge effect the perceived transparency. Three average luminance levels were chosen on the sine wave background: low luminance (LL), mid luminance (MS), and high luminance (HL) at 2 cd/m², 18 cd/m², and 40 cd/m² respectively, but with the same Michaelson contrast at 0.2. Four contrast levels were chosen on the sine wave background: low contrast (LC), mid contrast (MS), high contrast (HC), and very high contrast (HHC) with respective contrast at 0.05, 0.2, 0.4, and 0.93, but the same luminance at 18 cd/m². The background MS served as both the mid luminance and the mid contrast condition. An additional square wave grating (MQ) was chosen with the same contrast and average luminance as sine wave MS. The background specification was from the observer's position, meaning the reflectance and transmittance of the beam splitter were already accounted for in the values provided. The background gratings had the spatial frequency at 3 cycles per degree (cpd) which is at human visual system peak sensitivity [17]. The background grating covered 6°x15° FoV representing the extended parafoveal field. The grating was surrounded by medium gray at 20% of the display white luminance and neutral scales for reference. All 7 backgrounds were included in Experiment 1. 3 backgrounds of different contrasts were selected in Experiment 2 (LC, MS, HC) for verification. Sample scenes can be found in each experiment section.

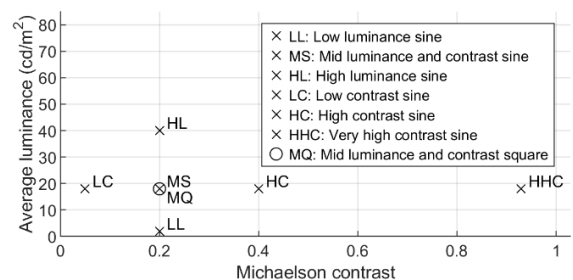


Figure 4. Specification of the 7 background gratings from the observer's position. Among the 6 sine waves, 3 were at different luminance level but the same contrast: LL, MS, HL, 4 were at different contrast levels with the same average luminance: LC, MS, HC, HHC. 1 square wave MQ had the same contrast and luminance as the sine wave MS.

"Test patches were 2°x2° FoV squares on the center area of the backgrounds to prioritize foveal vision. There were two types of test patches. One type reduced the contrast by increasing the

overall luminance through AR (AR patch), as along positive y axis inside the Figure 2 red rectangle. And the other from the background LCD (LCD patch) decreased the peak-to-peak luminance difference along negative x axis in Figure 2 blue rectangle. Here transparency was defined as the characteristics of a layer which can be seen through to the background. If it does not appear to have a layer between the observer and the background, it is categorized as “completely transparent”. If the background is not visible through the layer, it is categorized as “completely opaque”. Physical samples with various transmittance decreasing transparency towards either black or white opacity were presented to the observer to familiarize with the concept of perceived transparency. The observer had a training session to practice before each experiment which contained 5 randomly selected trials from the main experiment.

2.3 Experiment 1: Transparency Scaling

All 7 backgrounds were examined in the scaling experiment. Test patches in both types at 10 contrast levels were examined with each background. Each contrast was repeated for 3 times. In total, 2 patch types x 7 backgrounds x 10 contrast levels x 3 repeat resulted 420 trials each observer. During the experiment, the observer first adapted to one test background for 10s. A test patch from either AR or LCD type was presented randomly in a trial as constant stimulus, appearing in a random location within a $2^\circ \times 8^\circ$ area at the center of the background to minimize afterimage of the background pattern. The observer was asked to “rate the patch transparency by moving the slider with a mouse between the two anchors (completely transparent and completely opaque)” as seen in Figure 5 left subfigure. Observers had unlimited time to assess a patch before making decisions. All test patches were examined on this background before moving to the next background, and each patch was repeated three times. The experiment was separated into two sessions: 5 backgrounds (LL, MS, HL, LC, HC) in the first session, and 2 backgrounds (HHC, MQ) in the second session. 20 observers with normal or correct to normal vision participated in the experiment (15 males and 5 females). Observers both with and without experience with mixed reality were included. All observers participated in the first session for the 5 backgrounds. 5 observers returned and participated in the second session for the additional 2 backgrounds.

2.4 Experiment 2: Equivalent Transparency

The purpose of this experiment is to check if there is a perceived equivalency in transparency between the two types of patches (AR and LCD) through direct comparison. Though they could be rated at the same level of transparency in the scaling experiment, it was not clear if two types of patches were evaluated with the same criterion.

The experiment used paired comparison with an interleaved multiple simple staircases method. Three backgrounds (LC, MS, HC) of different contrasts were used in this experiment. LCD patches were used as the reference at 7 contrast levels relative to its background at 0.078, 0.204, 0.257, 0.311, 0.369, 0.4343, 0.636, respectively. Individual reference patch contrast was dependent on the background contrast. Each reference contrast of each background was associated with one staircase. Observers adapted to a background for 10s first. In each trial, one of the seven staircases were randomly selected. Two patches were displayed on the background: the reference LCD patch and the test AR patch (Figure 5 right subfigure). Both patches were side by side with 6° FoV separation, with their positions on the left or the right randomized. Observers were asked to select the patch that appeared more transparent to them using the arrow keys. The initial AR patch contrast was either minimum possible in our

setup, or maximum as the background contrast without additional AR overlay. One staircase would terminate when seven reversal points occurred or maximum limit of 25 trials was reached. Observers finished all seven staircases before moving to the next background. 10 observers with normal or correct to normal vision participated in the experiment (8 males and 2 females).

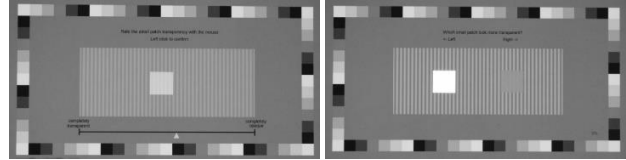


Figure 5. Photograph of example experiment interfaces. Left: Experiment 1 with MS background. The patch was from AR and the observers were asked to rate the transparency of the small patch in the middle using the slider. Right: Experiment 2 with HC background. The left patch was from AR and the right patch was from LCD. The observers were asked to select which of the square patches was more transparent.

3 RESULT

3.1 Transparency Scaling

Observers' transparency ratings were converted into numerical scales from 0 to 1, with “completely opaque” as 0, and “completely transparent” as 1. First, we examine the observer's rating consistency on the 3 repetitive trials to exclude observer's mistakes. A trial is discarded if it meets both requirements: 1. Its median absolute deviation is larger than three. 2. Its standard deviation (SD) is larger than a random distribution standard deviation (0.288). The outlier detection test removed 1.0% trials in AR test patch group, and 3.6% trials in LCD group. On average, observers took 3.6s to assess the appearance before giving response with SD of 4.3s. On both types of patches, an increase in background contrast resulted in increased average response time (LC 3.5s, HC 3.8s, HHC 5.7s). On AR patches, increased background luminance also increased the average response time (LL 3.2s, MS 3.5s, HL 3.8s), but this trend was not clear in LCD patches.

We pooled observers' responses, fitted them with normal distributions using maximum likelihood estimation, and used μ in the fitted distribution as the average rating [18]. The standard error of the estimated rating was calculated from the variance-covariance matrix of the estimator. The estimated standard error of the mean (SEM) on the background HHC and MQ are significantly higher than other backgrounds due to the smaller observer population. The estimated SEM was smaller on both the high and low contrast ends compared to the middle.

We normalized the test patch contrast to the background contrast. The average estimated transparency ratings with 95% confidence interval (CI) error bars were plotted against the test patch relative contrasts in Figure 6. The transparency rating curves for the AR patches are always different in shape from the rating curve for the LCD patches, which demonstrated that reducing contrast by the two methods activates different mechanisms in transparency perception. On backgrounds with the same contrast but different luminance levels (top two plots), the transparency ratings were not different from each other on either AR patch or LCD patch as their 95% CI bars overlap. The transparency ratings were similar on sinusoidal wave and square wave at same luminance and contrast (middle two plots). Looking at the bottom two plots, it is clear that for the AR patches, the perceived transparency on various normalized contrasts was still affected by the background contrast as their 95% CI bars do not

overlap. But on the LCD patches, background contrast did not impact the perceived transparency significantly except a small deviation on the highest contrast background.

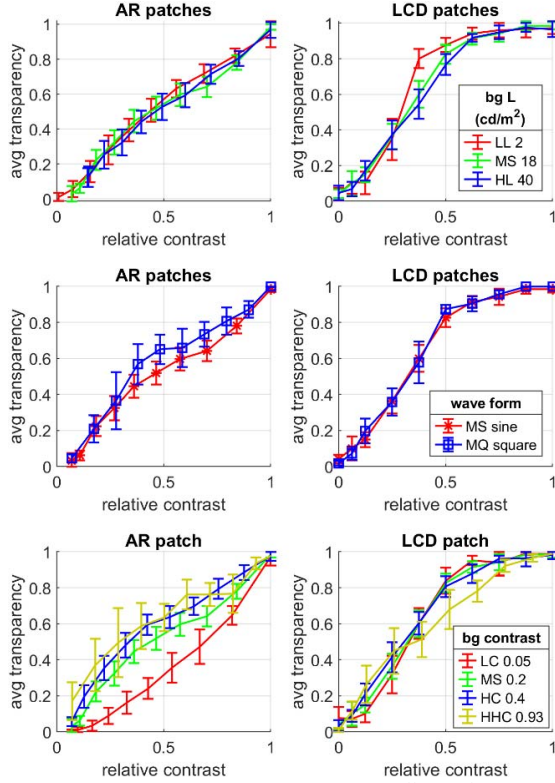


Figure 6. Experiment 1 result. Top row: effect of background average luminance on the transparency ratings. Middle row: effect of background wave form on the transparency ratings. Bottom row: effect of background contrast on the transparency ratings. The left and the right column shows AR patch and LCD patches respectively. The error bars represent the 95% CI of the mean.

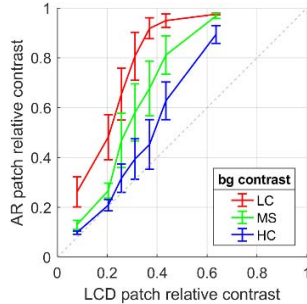


Figure 7. Experiment 2 result on equivalent transparency between AR patches and LCD patches on different contrast backgrounds (LC, MS, HC). The error bars represent the 95% CI of the mean.

3.2 Equivalent Transparency

All staircases for all observers terminated with seven reversal points before reaching the trial limit. Some observers reported difficulty making decisions on low contrast test patches and low contrast backgrounds. Observers spent 5.4s on average comparing the patch transparency with median of 3.5s and SD of 6.9s. The matched contrast was again pooled and estimated with maximum likelihood. We plotted the relative contrast of equivalent transparency between the reference LCD patches and the test AR

patches in Figure 7 with error bars representing 95% CI. Referring to the lower-left pane of Figure 6, the deviation from the diagonal dash line on all three backgrounds supports the separated mechanisms between two types of patches in Experiment 1. The separation of the three curves confirmed the effect of the background contrast on the perceived transparency.

4 MODEL TRANSPARENCY FROM CONTRAST REDUCTION

4.1 Contrast based transparency model

We proposed a model predicting the perceived transparency based on the patch type and background contrast based on the result from Experiment 1. On the AR patches, the perceived transparency is described with exponential curves as a function of the normalized contrast to the background. The perceived transparency T on the AR patches can be described as

$$T = |C/C_b - a|^b \quad (2)$$

, where C is the Michelson contrast of the additive patch area, and C_b is the Michelson contrast of the background. The parameters a and b were optimized with least square fitting to the experiment measured transparency data where a is a constant and b is a reciprocal function of the background contrast:

$$a = 0.0592 \quad (3)$$

$$b = 0.05737/C_b + 0.4315 \quad (4)$$

Since the additive AR overlay does not affect the peak-to-peak luminance difference, the term C/C_b can also be written as in the average luminance of the background $\bar{L}_b$ and the additive AR luminance $\bar{L}_{AR}$ as

$$C/C_b = \bar{L}_b/(\bar{L}_{AR} + \bar{L}_b) \quad (5)$$

On the LCD patches, All backgrounds results from Experiment 1 were pooled to optimize the model parameters. The transparency curve is modeled with single cumulative Weibull function described as:

$$T = 1 - e^{-(2.5C/C_b)^{1.75} - 0.05} \quad (6)$$

In the LCD patch condition since the average luminance $\bar{L}$ on the test patch is the same as the background, the term C/C_b can also be written as peak-to-peak luminance difference ratio of the patch ΔL and the background ΔL_b :

$$C/C_b = \Delta L/\Delta L_b \quad (7)$$

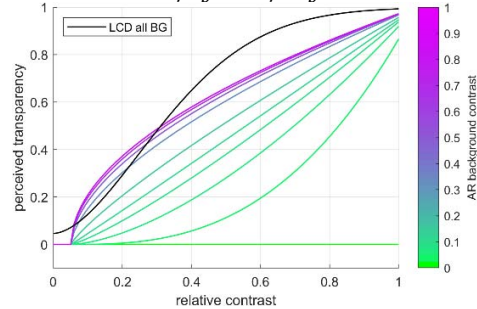


Figure 8. Proposed model prediction on the perceived transparency for different AR background contrast levels (see color scale) and LCD patches for any background (black).

With the modeled transparency, the discrete patch contrast and background contrast levels in the experiment can be extended to continuous values using Eq.(2-4) and (6). A collection of background contrast curves on AR patches are plotted in Figure 8 from green to purple colors. The LCD transparency is plotted in black and independent from the background contrast. All AR curves intercepted the x axis at relative contrast at 0.05, below which the patches are perceived to be completely opaque, indicating AR patch perceptual opacity is possible even with the

constantly presented backgrounds. Curves intercept with the full background contrast at different transparency levels ranging from 0 (opaque on solid background) to 0.97 (on maximum contrast background where there is physically no addition of light from the AR overlay). With the maximum background contrast, the curve has the exponential index of 0.51. With a solid background at 0 contrast, the patch will also be solid and perceived to be completely opaque regardless of the luminance of the additional AR overlay. On the LCD patches, the transparency ranges from 0.05 to 0.91 while the normalized contrast ranges from 0 to 1, which indicates the LCD patches do not appear to be completely transparent or completely opaque in any relative contrast levels. The transparency rating being larger than 0 when the LCD patch is indeed a solid patch (0 relative contrast) with patterned surround representing the effect of the pattern afterimage on the test patch appearance. The curve shape difference between the two types of test patches indicates they trigger different mechanisms in transparency perception.

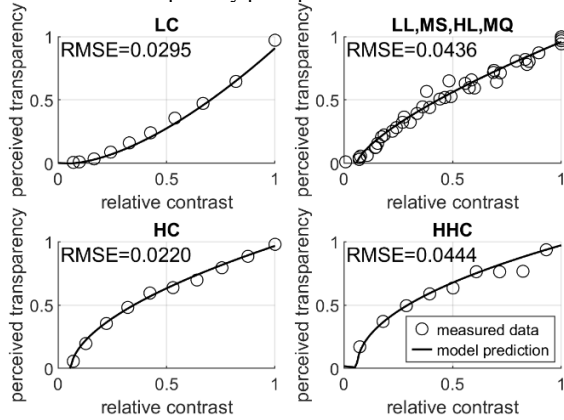


Figure 9. Model performance on perceived transparency with AR patches compared to Experiment 1 result. 4 background contrast levels are shown while background LL, MS, HL, and MQ at the same contrast and pooled to show in the top right subfigure.

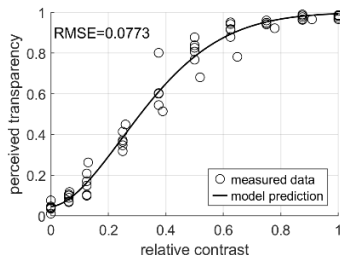


Figure 10. Model performance on perceived transparency with LCD patches compared to Experiment 1 result. All backgrounds are pooled due to the model independency to the background absolute contrast.

4.2 Model verification

The model prediction compared to the average experiment result is illustrated in Figure 9 for the AR patches, and Figure 10 for the LCD patches. For the AR patches, since the model is only contrast dependent and there are 4 backgrounds have the same contrast (LL, MS, HL, MQ), the results of the 4 backgrounds were pooled for the comparison in Figure 9 top right subfigure. The model performance is evaluated with the average root mean squared error (RMSE) at 0.0349. The largest RMSE is on HHL background at 0.0444. On the LCD patch, experiment results from all backgrounds are pooled for the evaluation. The model prediction RMSE is 0.0733.

We also applied the transparency model to predict the equivalent transparency in Experiment 2. The comparison between the model-estimated AR contrast against the measured is shown in the Figure 11. The model shows a good match to the experiment data with slight overall underestimation on the AR patch contrast in MS background (plotted in green). The correlated coefficient R^2 on the measured and predicted AR contrast are 0.9703, 0.9843, and 0.9969 on the background LC, MS, HC respectively. RMSE of the model prediction are 0.0041, 0.0210, 0.0120 on the background LC, MS, HC respectively.

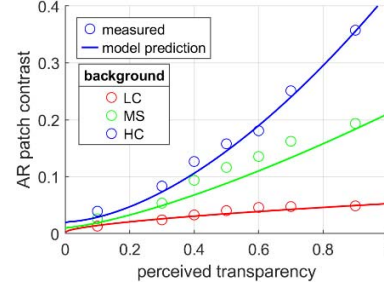


Figure 11. Model performance on equivalent transparency compared to Experiment 2 result. 3 tested background with different contrast levels (LC, MS, HC) are shown.

4.3 State-of-art transparency models

In the simulated transparency study by Singh and Anderson [8], they matched perceived transmittance at different average luminance levels by adjusting peak-to-peak luminance difference. Only high contrast (0.945) sinusoidal grating background stereoscopic test patches at a nearer disparity were. The result of the three observers showed that the transmittance matches when the patch Michelson contrast. The perceived transmittance was modeled linearly to the relative contrast to the background. Our result on the AR patches showed that in certain background contrast, the perceived transparency can be linear to the test patch contrast. But this linear relation can be altered with the background contrast. It is also debatable that if the perceived transparency in our experiment is equivalent to the perceived transmittance in their experiment.

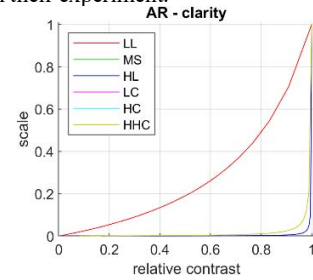


Figure 12. Predicted clarity on AR patches from subtractive filter transparency model [12]. Background LC, MS, HC, HHC overlap with each other.

In the subtractive filter appearance model by Faul and Ekroll [11], [12], [19], linear perceptual transmittance and clarity (perceived haziness) were modeled based on the physical transmittance and reflectance. The predicted transmittance stays constant at 1 regardless of the background conditions in our experiment setup, so is the clarity prediction on the LCD patch. The clarity prediction on the AR patch (Figure 12) does not fully predict the results in our experiment (see Figure 6 lower-left subfigure). The predicted clarity on LC, MS, HC, HHC overlap with each other and do not reflect the impact of background contrast on transparency.

5 DISCUSSION

Perceived transparency is not generally incorporated in color appearance models, but it is modulated by the AR luminance, which is conventionally evaluated in luminance or lightness. A comparison between linear transparency, luminance and lightness is illustrated in Figure 13. Since the modeled transparency is independent from background luminance as the experiment result indicates, luminance range is scaled down to illustrate the additional light from AR with sinusoidal pattern with contrast of 0.4 as the background. The first column from the top to the bottom shows equal steps of linear additional luminance from 10% to 90% maximum additional luminance. The second column shows equal steps of lightness from 10 to 90 in CIELAB style with the maximum luminance as reference white. The third and the fourth columns show steps of the linear transparency scale from 0.9-0.1 from the proposed model prediction, corresponding to the AR transparency and LCD transparency respectively. This figure is for illustration purpose and valid only under specified viewing condition on calibrated displays (letter width paper, 40 cm viewing distance in dark room, sRGB gamma with peak luminance at 150 cd/m²). It demonstrates that the transparency scale in OST AR is indeed unique and cannot be accurately estimated with existing dimensions like luminance or lightness, which highlights the need of incorporating the new perceptual dimension in AR color management pipelines to improve image quality.

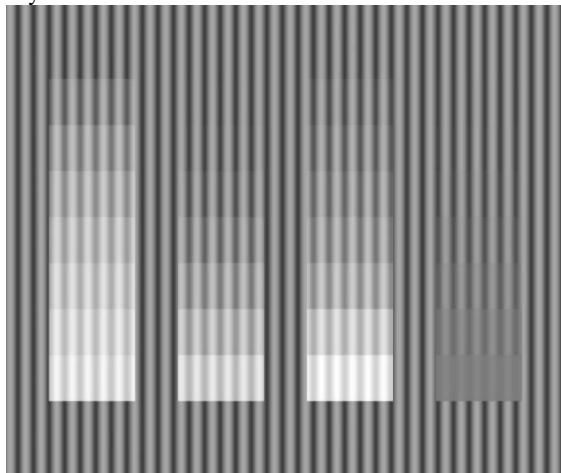


Figure 13. Transparency scale compared to luminance and lightness. From left to right: linearly increased luminance, linearly increased lightness, rendered linearly decreased AR transparency, rendered linearly decreased LCD transparency. This illustration is only valid on calibrated displays and viewing conditions.

6 CONCLUSION

We measured perceived transparency from contrast modulation with two experiments on patterned backgrounds. The experiments included contrast manipulation on average luminance from AR overlay, and peak-to-peak luminance difference from simulation. The first experiment measured the perceived transparency scale directly and the second measured the equivalent transparency between the two types of contrast manipulation. The results of both experiments showed fundamental differences in perceived transparency between the two types of contrast modulation. On the AR overlays, the background contrast and the relative contrast to the background impact the perceived transparency while background luminance and wave form do not. On the simulated patches, only the relative contrast affects the perceived

transparency. We modeled this background contrast related transparency according to the relative contrast, the background contrast, and the modulation type from the first experiment result. The model was verified with the second experiment and showed good prediction with average R^2 at 0.9838.

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